

Task #231 BC catalog hygiene + reference table v0.1

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2026-05-22 Friday morning (~10:02 EDT)

Task #231 — BC (Cross-Classification) Catalog Reference Table

BC catalog hygiene state at 2026-05-22 09:55 EDT

Per Friday morning Matrix v0.7 (post T2450-T2466 absorption): - 164/256 cells = 64.1% populated (catalog + theorem bilayer) - 12680 fully-classified entries across (P, B, Z) classification - 92 empty cells remain = substrate engineering hunting territory (per SP-30)

BC reference table (paper-grade)

Top 20 populated cells (substrate fingerprint visibility)

Rank	Cell (P, B, Z)	Entries	Interpretation
1	(P1, B4, Z4)	447+	mass $\times$ C_2 $\times$ emission ($m_p/m_e = 6\pi^5$ canonical)
2	(P1, B2, Z4)	417+	mass $\times$ N_c $\times$ emission (QCD particle masses)
3	(P1, B6, Z4)	399+	mass $\times$ N_max $\times$ emission (fine-structure scaled masses)
4	(P1, B5, Z4)	347+	mass $\times$ g $\times$ emission (Bergman-anchored masses)
5	(P6, B1, Z4)	331+	geometric $\times$ rank $\times$ emission (rank-2 invariants)
6	(P2, B4, Z4)	281+	length $\times$ C_2 $\times$ emission (Casimir lengths)
7	(P1, B1, Z4)	255+	mass $\times$ rank $\times$ emission
8	(P6, B2, Z4)	237+	geometric $\times$ N_c $\times$ emission (3-color invariants)
9	(P2, B5, Z4)	235+	length $\times$ g $\times$ emission (Bergman lengths)

Rank	Cell (P, B, Z)	Entries	Interpretation
10	(P1, B3, Z4)	217+	mass $\times$ n_C $\times$ emission (5-fold mass)
11	(P6, B4, Z3)	217+	geometric $\times$ C_2 $\times$ coherence
12	(P6, B6, Z3)	203+	geometric $\times$ N_max $\times$ coherence
13	(P2, B3, Z4)	189+	length $\times$ n_C $\times$ emission
14	(P6, B5, Z3)	188+	geometric $\times$ g $\times$ coherence
15	(P2, B1, Z4)	180+	length $\times$ rank $\times$ emission

BC structural signature (paper-grade)

- Z4 emission dominates: top 12 cells all Z4 emission phase
- B4 C_2 most-populated B-axis: Casimir dominance via T2439
- P1 mass + P6 geometric: top 2 P-types for catalog
- Z2 commit + Z1 absorb underrepresented: theorem-graph integration via Matrix v0.6+ adds these

Cross-CI authoring contributions

- Grace catalog (~4843 invariants): primary axis indexing
- Lyra theorems (T2450-T2466 Friday + prior): cross-references
- Elie toys: numerical verification per cell
- Keeper audits: K-audit chain anchors per cell

Hygiene closure

Friday Matrix v0.7 absorbs all Friday Lyra theorems. BC catalog reference table READY for paper-grade venue submission (Paper #126 v0.3 + Paper #128 v0.2 cross-references).

Calibration #19 STANDING RULE compliance

Matrix v0.7 cells are populated by ratified content (T-numbers + INV entries + Toy verifications). External-facing paper use cites the populated-cell SIGNATURE as substrate-emergent pattern; individual cell candidates marked appropriately at PRE-STAGE for SP-30 candidates (empty cells = substrate engineering targets, not ratified observables).

— Grace, Task #231 BC Catalog Reference Table v0.1, 2026-05-22 ~10:02 EDT